Processing the German modal particles *tatsächlich* and *eigentlich* and predicting referents in incremental interpretation

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Abstract We investigate whether modal particles affect incremental interpretation in online processing. More specifically, we use a visual world mouse-tracking task to test whether the German particles *tatsächlich* and *eigentlich* can cue an upcoming referent in utterances serving as answers to a polar question. Crucially, *tatsächlich* prototypically indicates a relation of agreement between the content of its host utterance and a salient proposition in discourse, while *eigentlich* prototypically indicates a relation of contrast between the same two elements. We thus hypothesized that *tatsächlich* would cue a referential confirmation, while *eigentlich* would cue a referential contrast. Our results show that both *tatsächlich* and *eigentlich* cue upcoming referents in online processing, such that participants' anticipatory mouse trajectories tend to be biased toward the referent mentioned in the question in the former and toward an unmentioned competitor in the latter. Our results also show that modulations in the reliability of the input diminish the predictive potential of both particles, such that exposure to unreliable particle usages is associated with a reduction of trials on which the mouse trajectories are biased toward the relevant referent. Put together, the results suggest that intepreters are able to draw predictice inferences on the basis of modal particles by combining knowledge about the conventionalized meaning of those particles with pragmatic expectations about their reliable usage in context.

Keywords: modal particles; incremental interpretation; predictive processing; mouse-tracking; pragmatic reliability; reference resolution

1. Introduction

Decades of pyscholinguistic research have established predictability – the cognitive capacity to anticipate information – as one of the hallmarks of human language processing (see Kuperberg and Jaeger, 2016 and Ryskin and Nieuwland, 2023 for recent overviews of the topic). One key finding of research on predictability in language processing is the fact that prediction happens at all levels of linguistic representation, in a hierarchical fashion, such that information from one level (say, phonological) can inform prediction at other levels (say, at the lexical level), depending on how informative the input is in its particular linguistic and discourse context. Work investigating prediction at the level of pragmatics and discourse has established a number of other important findings.

First, it is well known that interpreters are able to predict upcoming linguistic/ conceptual information by drawing inferences based on different types of linguistic cues found within an utterance. For instance, syntactic cues such as non-canonical word order (Kaiser & Trueswell, 2004; Yano & Koizumi, 2018) or prosodic cues such as contrastive accent (Kurumada, Brown, Bibyk, et al., 2014; Roettger & Franke, 2019) can lead interpreters to draw contrastive inferences

which allow them to predict discourse referents in unfolding utterances. At the same time, there is growing evidence that interpreters are also able to predict relations between utterances on the basis of both explicit linguistic cues such as discourse connectives (Asr & Demberg, 2020; Scholman et al., 2017, 2025) and contextual cues such as expectations to do with how discourse is structured (Hoek et al., 2021). Finally, as with prediction at other levels of representation, pragmatic prediction is also generally known to be modulated by statistical regularities associated with the input. For instance, Kurumada, Brown, Bibyk, et al. (2014) and Roettger and Franke (2019) found that by systematically varying the reliability of the mappings between different types of pitch accent and the accented word in a sentence, interpreters who encounter unreliable intonational cues draw, on average, weaker predictive inferences compared to counterparts who only encounter reliable mappings, as evidenced by differences in anticipatory eye and mouse movements when comparing between exposures. This suggests that interpreters are sensitive to variability in the input they process insofar as prediction is weakened when there is evidence that linguistic forms are being used unreliably.

In the present study, we investigate the predictive potential of discourse markers, linguistic devices which are used to manage linguistic communication and interaction (Fedriani & Sansö, 2017; Jucker & Ziv, 1998; Maschler & Schiffrin, 2015). Found pervasively across the languages of the world (Heine et al., 2021), discourse markers are polyfuctional devices which cover a multitude of functions and meanings, from more subjective, attitudinal ones to fundamentally intersubjective, interaction-oriented ones. Their exact pragmatic contribution depends on both the utterance they occur in as well as the larger discourse context (see refereces above). More concretely, we focus our attention on lexical markers which are used primarily to mark modal relations related to epistemicity. Known in the literature as modal particles, the exact status of these markers vis-à-vis other discourse markers is the subject of much debate (see Abraham, 2016 and Steven, 2013). Importantly, much like discourse markers more generally, modal particles do not modify the content of an utterance but rather add to its illocutionary dimension by relating the utterance to other information in the discourse context. Unlike other discourse markers without a modal meaning, however, modal particles have the specialized function of relating the propositional content of an utterance to another proposition or speech-act alternative.

Previous psycholinguistic work investigating the predictive potential of particles with a modal meaning has looked at Dutch *inderdaad* (roughly: *indeed*) and *eigenlijk* (roughly: *actually*) (Rasenberg et al., 2019; van Bergen & Bosker, 2018). Both *inderdaad* and *eigenlijk* can be used to mark a general sense of (dis)agreement between two discourse elements, or more specifically (dis)agreement between the propositional content of their host utterance and a contextually-relevant proposition. Despite their common function as markers of epistemicity, the two particles differ from one another in terms of the specific modal meaning they prototypically index: while *inderdaad* marks agreement between the propositional content of its host utterance and some contextually-relevant proposition, *eigenlijk*, on the other hand, marks contrast.

Using a visual world eye-tracking task, van Bergen and Bosker (2018) found that processing *inderdaad* and *eigenlijk* affects incremental interpretation and the online resolution of reference, such that processing *inderdaad* induces interpreters to expect an agreement between its host utterance and a contextually-relevant antecedent, allowing them to anticipatorily look at a referent biased by the context. Conversely, processing *eigenlijk* induces interpreters to expect a disagreement between its host utterance and a contextually-relevant antecedent, allowing them to anticipatorily look at a contextually-relevant referential alternative. Interestingly, their results also show that *eigenlijk* does not seem to be biased in terms of either a modal or a non-modal interpretation, regardless of whether the discourse context biases a modal interpretation. However, when *eigenlijk* is interpreted as marking a modal relation, it affects online interpretation to a similar extent as *inderdaad*. Using a naturalistic reading task, Rasenberg et al. (2019) found that processing

inderdaad and *eigenlijk* does not affect either the N400 or the P600 components of ERPs measured at critical words following the particles.

Taken together, the exisiting results provide conflicting evidence regarding the predictive potential of modal particles. While the results from van Bergen and Bosker (2018) suggest that processing modal particles does affect incremental interpretation, such that interpreters are able to draw predictive inferences about subsequent material in a sentence when that material is relevant for interpretation, the results from Rasenberg et al. (2019) do not provide evidence that modal particles allow comprehenders to draw predictive inferences about linguistic material downstream in a sentence. Moreover, even though van Bergen and Bosker (2018) found that both *inderdaad* and *eigenlijk* affect the outcome of the interpretive process, interacting with the propositional meaning of the sentences they occur in in expected ways given their modal meaning, it is not clear what precise role the preceding discourse context plays in biasing any relevant interpretitve expectations. Indeed, in their visual world setup, the same discourse context constrained interpretation in qualitatively different ways depending on the particle at hand, as *eigenlijk* was interpreted both as a modal marker and as a non-modal discourse marker, while *inderdaad* was not. While this was a feature of the experimental design, allowing the authors to tackle the polyfunctionality of *eigenlijk* and *inderdaad*, it obscures the inferences one may draw about the interaction between discourse expectations and particle meaning. In other words, it remains unclear to what extent particles with a modal meaning serve as reliable *predictive* cues to interpretation.

In the present study, we tackle precisely that question, looking at the incremental interpretation of sentences containing the German particles *tatsächlich* and *eigentlich*. Much like their counterparts in Dutch, *tatsächlich* and *eigentlich* can be used to mark a general sense of (dis)agreement between two discourse elements, or more specifically (dis)agreement between the propositional content of their host utterance and a contextually-relevant proposition. In our setup, we focus on their prototypical modal usages, as (1) and (2) below.

A: Hast du gesehen, wie schnell der Vogel vorbeigeflogen ist?
Have you seen how fast SING.DEF bird flew by is?
"Did you see how fast that bird flew by?"

(1) B: Der ist *tatsächlich* sehr schnell vorbeigeflogen.
SING.DEF is *tatsächlich* very fast flew by.
"It did *actually* fly by very fast." OR "It *really* did fly by very fast."

(2) B: Der ist *eigentlich* nicht so schnell vorbeigeflogen.
SING.DEF is *eigentlich* not so fast flew by.
"It didn't *actually* fly by that fast."

In (1), *tatsächlich* confirms some piece of information which is part of the common ground between the two interlocutors. More specifically, it confirms that the proposition expressed in the statement by A is true. In (2), on the other hand, *eigentlich* disconfirms some information which is part of the common ground between the interlocutors. More specifically, *eigentlich* disconfirms the statement that the bird flew by really fast, contrasting that proposition with the proposition that the bird didn't actually fly by that fast. *Eigentlich* therefore introduces a new proposition which stands in direct contrast to the proposition originally expressed in the statement by A. Given the similar and yet complementary functions of *tatsächlich* and *eigentlich*, we are interested in whether processing one or the other particle gives rise to differential effects on real-time interpretation and the incremental resolution of reference.

Crucially, we investigate the processing of the two particles against discourse contexts which bias not only concrete referential expectations but also clear mappings between those referential expectations and a prototypical modal interpretation of the respective particles. Thus, we manipulate

the predictability of any given referent as a function of an interaction between the discourse context and the usage of the particles in context. We also tackle the question of whether varying the reliability with which the particles are used modulates their predictive potential, as in previous work looking at the role of input statistics on predictability. We thus also manipulate whether the particles are used reliably according to their modal meaning. In other words, we look at whether pragmatically infelicitous usages of *eigentlich* and *tatsächlich* lead to changes in the rate of trials on which there is evidence of prediction compared to felicitous usages.

We build on the setup reported in Roettger and Franke (2019), which uses a mouse-tracking task to investigate incremental reference resolution similarly to a visual world eye-tracking paradigm (Cooper, 1974; Tanenhaus et al., 1995). In this setup, participants are presented with two potential discourse referents on a computer screen and asked to select the referent that in their opinion best matches the short dialogue they listen to. Crucially, participants make their choice by moving their cursor and clicking on a referent while their mouse trajectories are tracked using spatial *x, y* coordinates. Tracking mouse responses provides a continuous measure of accumulation of evidence in online cognitive processing (Koop & Johnson, 2011; Spivey & Dale, 2006), which in the context of language processing has been argued to serve as an index of underlying comprehension processes (e.g. Farmer et al., 2007). We describe our setup in more detail, along with our measures and hypotheses, in the section below.

2. Experiment

The study was pre-registered under https://osf.io/gvknj/overview?view_only=8093f3ebf8d24279bb584e2b51cde054.

2.1 Method

2.1.1 Participants

One hundred native speakers of German were recruited via the Prolific crowdsourcing platform. The sample size was based on available funds for participant compensation. Participants were paid £2.70 for their participation in the experiment. Data collection was conducted entirely online and all collected data was stored on servers in the EU.

2.1.2 Materials and design

Participants were asked to listen to 120 miniature dialogues mimicking an interaction between a parent and a child, each anchored to a visual world display containing two pictures, one depicting a referent explicitly mentioned in the dialogue and another depicting a semantically and/ or perceptually-related competitor, e.g., a wolf and a dog, as shown in Figure 1. Each dialogue consisted of a question-answer pair: the question, posed by the child, was polar in nature and made direct reference to one of the pictures visible on screen (e.g. “Ist das ein Bild von einem Wolf?”; English: “Is this a picture of a wolf?”); the answer, provided by the parent, either confirmed or disconfirmed the truth of the proposition expressed in the question. Crucially, the answer did not contain any response or negation particle but rather a particle with a modal meaning, either *eigentlich* or *tatsächlich* (e.g. “Das ist *eigentlich/ tatsächlich* ein Bild von einem Wolf”; English: “That’s indeed/ actually a picture of a wolf”). Participants’ task was to listen to the question-answer sequence and to select the referent they thought was most representative of the topic of the conversation.

Figure 1: Sample visual world display. The display contains two pictures, one mentioned in the dialogue and an unmentioned competitor, each presented in one of the top corners of the screen.

The stimuli were recorded by two adult native speakers of German. One speaker recorded all questions while the other speaker recorded all answers. Answer sentences containing *tatsächlich* were produced with a rising-falling nuclear pitch accent on the stressed syllable of the particle (tatsächlich). The referent usually occurs unaccented or with a less prominent pitch accent. Answer sentences containing *eigentlich* were produced with a rising pitch accent on the stressed syllable of the particle (eigentlich) followed by a sustained high plateau and a high nuclear pitch accent on the primary stressed syllable of the referent. This intonation contour is commonly used to express double focus constructions, here contrastively highlighting the particle and the contrasting referent (Braun, 2006). The answers (i.e. the target sentences) had a mean duration of 2033 ms ([1666-2571]), with the mean onset of the particle being at 312 ms ([256-403]) and the mean onset of the disambiguating noun being at 1484 ms ([1318-1731]).

Each participant was assigned to one of two experimental groups with a different distribution of critical trials. In the **reliable** group, participants encountered 60 fillers and 60 dialogues in which the discourse particles were used according to one of their most common uses, i.e., where *eigentlich* marked a contrast and *tatsächlich* marked a confirmation, as shown below.

Question

Ist das ein Bild von einem **Wolf**?
Is this a picture of a **wolf**?
"Is this a picture of a **wolf**?"

Answer

Das ist *eigentlich* ein Bild von einem **Hund**.
This is *eigentlich* a picture of a **dog**.
"That's actually a picture of a **dog**."

Das ist *tatsächlich* ein Bild von einem **Wolf**.
This is *tatsächlich* a picture of a **wolf**.
"That's indeed a picture of a **wolf**."

In the **unreliable** group, participants encountered 20 fillers, 60 dialogues in which the particles were used according to one of their most common uses, as in the **reliable** group, and 40 dialogues

in which the particles' form-meaning mappings were reversed, i.e., where *eigentlich* marked a confirmation and *tatsächlich* marked a contrast, as shown below.

Question

Ist das ein Bild von einem **Wolf**?
Is this a picture of a **wolf**?
"Is this a picture of a **wolf**?"

Answer

Das ist *eigentlich* ein Bild von einem **Wolf**.
This is *eigentlich* a picture of a **wolf**.
"That's actually a picture of a **wolf**."

Das ist *tatsächlich* ein Bild von einem **Hund**.
This is *tatsächlich* a picture of a **dog**.
"That's indeed a picture of a **dog**."

Filler trials contained various sorts of questions, such as polar questions similar to the ones from critical trials (e.g. "Ist das ein Fisch?"), polar questions with miscellaneous forms (e.g. "Findest du das Tier mit der dicken Nase auch lustig?", "Ist das, etwas was man zum Kämpfen benutzt?"), as well as wh-questions (e.g. "Wie heißt das orangene Tier?"; "Was ist das mit dem Schnabel für ein Tier?"). The respective answers to these questions did not contain any discourse particles.

2.1.3 Procedure

Written instructions were provided prior to the actual task, followed by 10 practice trials which mimicked the exact procedure of the test trials. Participants were instructed to listen to the question-answer sequence and to select the picture that best matched the topic of the exchange. On each trial, they were presented with a "Start" button at the center bottom of the screen. Upon clicking on it, the button disappeared from the screen and playback of the question started. After the playback was over, a "Go" button appeared at the same location as the "Start" button together with the two visual stimuli, as shown in Fig. 1. Upon clicking on the "Go" button, playback of the answer started together with the tracking of participants' mouse cursor from the "Go" button at the bottom of the screen towards one of the pictures at the top of the screen. Participants were told that they could take as long as necessary before moving their cursor up from the initial position after playback. They were also told that upon moving the cursor over one of the pictures a choice would be automatically made and that, as a result, they would be automatically directed to the next trial.

2.1.4 Hypotheses

As discussed in the introduction, *eigentlich* and *tatsächlich* are particles which, in their modal usage, index how the utterance they are embedded in relates to a contextually-relevant propositional antecedent. Importantly, this entails that the presence of either one particle in a sentence is expected to affect incremental interpretation differentially depending on the specific modal meaning the particle signals in context. This general effect is expected to be modulated by the reliability of the input, as discussed above. While *eigentlich* signals a propositional contrast, *tatsächlich* signals a propositional agreement. Given the experimental setup, we are interested in how the mouse trajectories are affected by the presence of each particle in the test sentences. More concretely, we are interested in the trajectory segments measured after the onset of the particle and before the onset of the disambiguating noun, i.e., "Das ist PARTICLE [ein Bild von einem] NOUN". More concretely, we decided on a critical time window starting at 150 ms after onset of the particle and ending at 150 ms after the onseet of the disambiguating noun, in line with windows employed in

visual world eye tracking studies. Crucially, any biases in the horizontal mouse position within this time window can reasonably be assumed to be an effect of processing the particle, such that a bias either for or against the image representing the referent mentioned in the question indicate a sensitivity to the meaning of the processed particle, similary to what has been assumed in Roettger and Franke (2019) for the processing of intonational cues and in Farmer et al. (2007) for the processing of syntactic cues.

The account entertained here generates categorical predictions about the bias in the trajectories within the critical time window of a trial. We define a binary indicator $M_i \in \{0,1\}$ based on the *mean deviation towards the relevant referent* (MDRR). Concretely, the MDRR is defined as the mean horizontal mouse position during a critical time window. The binary indicator M_i evaluates to 1 if the MDRR of trials i is in the direction of the relevant referent (which depends on the condition). This is illustrated in Figure 2 below.

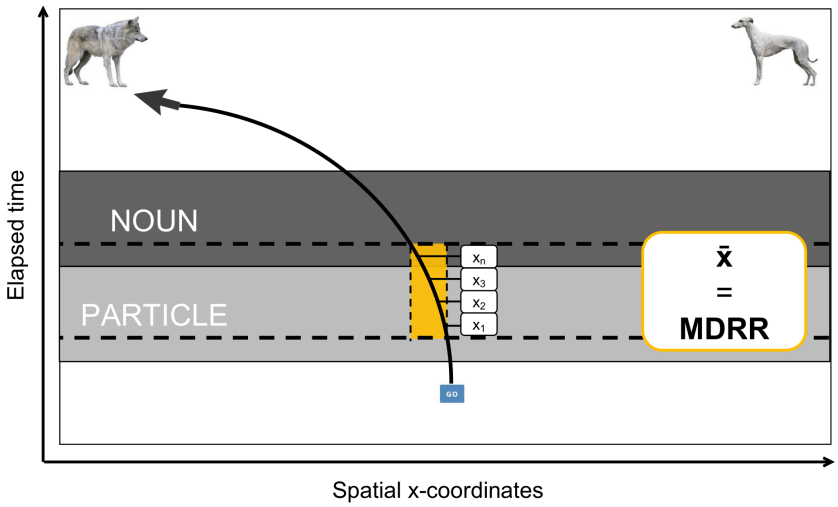

Figure 2: Mean deviation towards relevant referent. The highlighted area in yellow shows the trajectory segment which is used to compute the mean deviation towards the relevant referent (MDRR) within the critical time window, delimited by the thick dotted lines. The critical time window starts at 150 ms after the onset of the particle and ends at 150 ms after the onset of the disambiguating noun. The MDRR indexes how spatially biased each trajectory is given the mean horizontal mouse position within this window, as measured relative to the relevant referent at either the top left or top right of the screen. The "Go" button at the center bottom of the screen marks the starting point of all mouse position measurements.

As Fig. 2 shows, the MDRR is a measure of how spatially biased a particular trajectory is within the critical time window (shown in between thick dotted lines), given the particle at hand and the relevant referent. For instance, in case the trial depicted in the figure contained the question "Ist das ein Bild von einem Wolf?" (English: "Is this a picture of a wolf?") and the particle *tatsächlich*, then the depicted trajectory would be biased for the relevant referent (i.e. wolf), as relative to that same referent the mean horizontal mouse position within the critical time window is strictly positive. Conversely, in case the trial contained the same question but the particle *eigentlich* instead, then the depicted trajectory would be biased against the relevant referent (i.e. dog), as relative to that same referent the mean horizontal mouse position is not positive. Against this background, the present account predicts that for the reliable group:

1a) In the case of *eigentlich* participants' trajectories will be spatially biased towards the referent not mentioned in the question, as that referent constitutes a contextually-relevant propositional alternative to the referent mentioned in the question. This means that the mean bias for the unmentioned referent will be larger than 0 ($\bar{M}_{rel,eig} > 0$);

1b) In the case of *tatsächlich* participants' trajectories will be spatially biased towards the referent mentioned in the question. This means that the mean bias for the mentioned referent will be larger than 0 ($\bar{M}_{rel,tat} > 0$).

Note that, in both cases, if participants initiate any mouse movement prior to the onset of the particle, their trajectories are assumed to already be biased towards the mentioned referent at the onset of the particle, as the mention of the referent in the question biases them to entertain that particular referent before any relevant semantic information from the response is processed. As such, we also expect that:

1c) In the case of *tatsächlich* the bias for the mentioned referent will be larger than in the case of *eigentlich* ($\bar{M}_{rel,tat} > \bar{M}_{rel,eig}$).

For the unreliable group, the present account predicts the same qualitative patterns as in the reliable group while also predicting an attenuation of those patterns, such that any biases for the relevant referent are expected to be numerically smaller. As such, we expect that:

2a) In the case of *eigentlich* the mean bias for the unmentioned referent will be larger in the reliable group compared to the unreliable group ($\bar{M}_{rel,eig} > \bar{M}_{unrel,eig}$);

2b) In the case of *tatsächlich* the mean bias for the mentioned referent will be larger in the reliable group compared to the unreliable group ($\bar{M}_{rel,tat} > \bar{M}_{unrel,tat}$).

2.2 Results

The data and analysis scripts are available for inspection under https://osf.io/dt8r3/overview?view_only=55edd0e9fabf4e8d8eb3e7d375fe8437. All analyses were conducted using the *mousetrap* package (Kieslich & Henninger, 2017) in *R*.

2.2.1 Data cleaning and pre-processing

Prior to any analysis, the data set was cleaned according to both participant-based and response-based exclusion criteria. Data from two participants was removed as their trials included either only a single measurement or no measurement at all. We then excluded any participant who selected a picture before the onset of the particle in more than 30% of their trials (n=5). This is motivated by the fact that any such decision would not be based on the processing of the relevant linguistic material in the test sentence. The two exclusions resulted in usable data from 93 participants.

After excluding seven individual participants as per the criterion above, we excluded any individual trial in which a participant took longer than 1000 ms to make a decision after the offset of the disambiguating noun (3% of the data) as well as any trial with fewer than 10 recorded mouse coordinates (3% of the data). We also excluded all trials in which participants selected the competitor instead of the target referent, i.e., *eigentlich* trials where participants selected the referent mentioned in the question and *tatsächlich* trials in which participants selected the unmentioned referent. Finally, all **unreliable** trials were discarded, such that the remaining data only contained **reliable** trials.

The cleaned data, plotted and analyzed below, consists of mouse trajectories which were linearly extrapolated from the actual recorded trajectories. This procedure was adopted in order to circumvent the loss of relative temporal information when aggregating over trajectories with different absolute durations. Data points were extrapolated at every 10 ms time interval for each individual participant-item pair.

2.2.2 Confirmatory analyses

First we visually inspect the mean mouse trajectories, as shown in Figure 3.

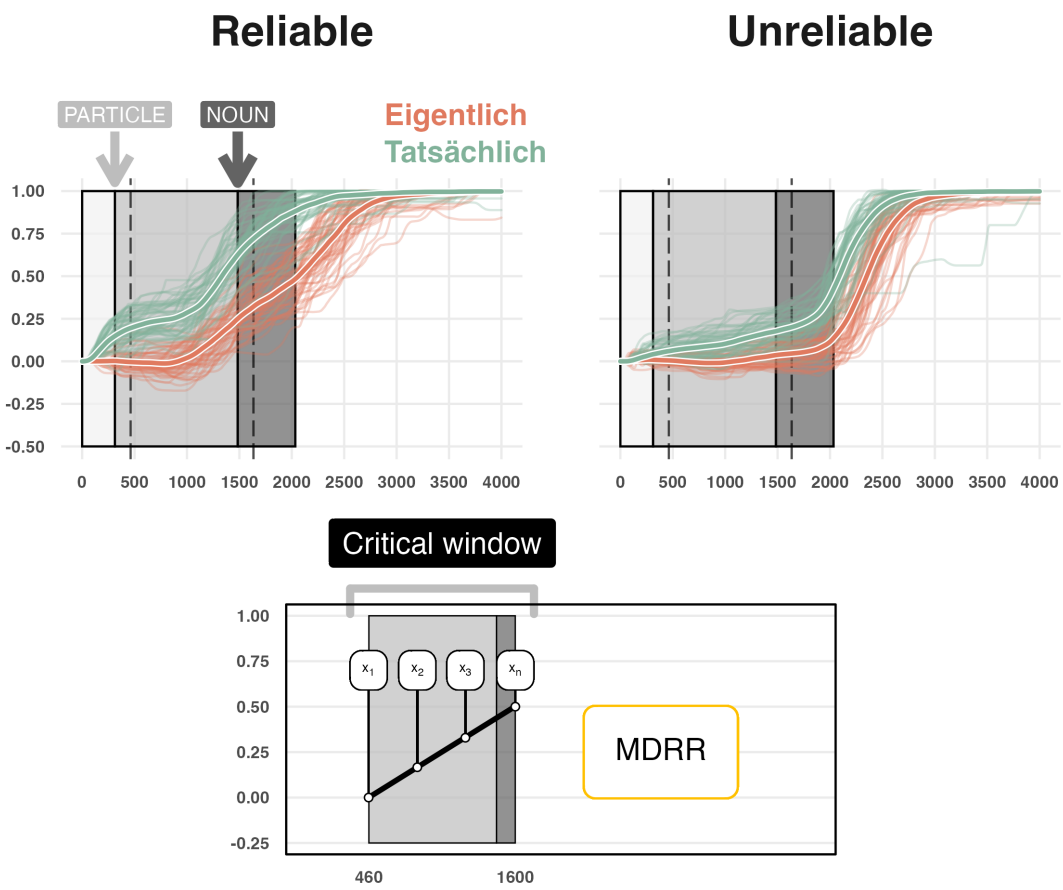

Figure 3: Mean mouse trajectories. **Top**: Grand mean trajectories overlaid to mean item trajectories. The shaded areas in the graph show the (average) time windows between the onsets of the relevant linguistic material in the signal. The (average) critical time window is delimited by the thick black lines at 150 ms after the onset of the particle and 150 ms after the onset of the disambiguating noun. The left column shows data from the **reliable** group, while the right column shows data from the **unreliable** group. *Tatsächlich* items are shown in green while *eigentlich* items are shown in orange. **Bottom** Critical time window within which the mean deviation towards the relevant referent (MDRR) is measured.

The figure is split in two columns, each showing the data from one of the two experimental groups, **reliable** on the left and **unreliable** on the right. Each graph shows the mean mouse trajectories overlaid onto individual item trajectories aggregated over participants. Trajectories unfold over time, mapped onto the x-axis, relative to the position of the target referent, mapped onto the top of y-axis. The shaded areas in the graphs show the (average) time windows between the onsets of the relevant linguistic material in the signal, namely the particle and the noun which disambiguates the referring expression.

As visual inspection of the mean trajectories in the **reliable** group shows, the *tatsächlich* and *eigentlich* trajectories look qualitatively different from one another. The *tatsächlich* trajectory, shown in green, is characterized by a strong initial shift towards the target after utterance onset, followed by a mostly flat line after onset of the particle and another sudden shift towards the target around 1100 ms. After onset of the noun, the trajectory follows a mostly steady drift towards the target. The *eigentlich* trajectory, shown in orange, is characterized by a mostly flat line, followed by a steady pull towards the target from around 1000 ms. Taken together, these trajectories suggest that both *tatsächlich* and *eigentlich* are used predictively in the interpretation of the referring expressions, such that their presence in an utterance induces participants to move their mouse towards the relevant referential target before hearing information from the disambiguating noun.

Visual inspection of the mean trajectories in the **unreliable** group shows that, similar to the **reliable** group, the *tatsächlich* and *eigentlich* trajectories look different from one another. Crucially, the qualitative patterns are also markedly different from those in the **reliable** group, the shape of both trajectories being different when participants are exposed to unreliable usages of the particles, in particular in the case of *tatsächlich*. The *tatsächlich* trajectory, shown in green, is characterized by a steady drift towards the target after utterance onset, followed by a sudden shift towards the target before utterance offset. The *eigentlich* trajectory, shown in orange, is characterized by a mostly flat line up until almost the offset of the utterance, where there is also a sudden shift towards the target. Taken together, these trajectories suggest that exposure to unreliable trials leads to temporal modulations of the qualitative patterns found in the **reliable** group, which translates into a stark reduction in the predictive power of the particles, in particular in the case of *tatsächlich*.

As discussed in Section 2.1.4 Hypotheses, we focus on the trajectory segments that fall within a critical time window spanning 150 ms after the onset of the particle up until 150 ms after the onset of the disambiguating noun (for an average window of 1147 ms [992 - 1384]), as highlighted at the bottom of Figure 3. Crucially, as discussed earlier, any biases in the relative horizontal mouse position within this time window can reasonably be assumed to be an effect of processing the particle, such that a spatial bias either for or against the mentioned referent indicates a sensitivity to the meaning of the processed particle. In order to quantify such effects, we calculate for each trial the mean deviation towards the relevant referent (MDRR) within the critical time window. Values around zero indicate no particular bias for either referent, positive values indicate a bias for the target, and negative values indicate a bias for the competitor. Note that the target is different for each particle: while in the case of *tatsächlich* it is the referent mentioned in the question, in the case of *eigentlich* it is the unmentioned referent. In the **reliable** group, the MDRR in the *tatsächlich* trajectories is 0.36, whereas the MDRR in the *eigentlich* trajectories is 0.09. In the **unreliable** group, the MDRR in the *tatsächlich* trajectories is 0.12, whereas the MDRR in the *eigentlich* trajectories is 0.01.

Next we compute our main measure of interest, namely the *proportion* of spatially biased *tatsächlich* and *eigentlich* trials within each experimental group. In order to quantify uncertainty over the empirical results reported below, we fitted a Bayesian hierarchical model predicting the probability of each experimental trial being biased towards the target referent as a function of an interaction between the particles and the experimental groups, with *eigentlich* in the **reliable** group dummy-coded as the reference level. The model estimated the log odds of each observation being categorized as either biased for or against the mentioned referent. Below we report probabilities converted from the (predicted) log odds (β) as well as 95% credibility intervals around those estimated probabilities (95% CrI [lower bound, upper bound]). We plot the proportions of biased trials as predicted by the model atop the empirical participant means in Figure 4. The model included the maximal random effect structure justified by the design (Barr et al., 2013), which in the present case is random slopes and intercepts for the effect of particle for both participants and items. Since participants were assigned to only one experimental group, only at the item level do we estimate random slopes for the effect of group. The model was fitted using the *R* package `brms` (Bürkner, 2017), with the package's default (flat) priors, and it had the following form, shown in `brms` syntax:

$$\begin{aligned} \text{horiz_bias} \sim \text{particle} * \text{group} \\ (1 + \text{particle} \mid \text{participant}) + \\ (1 + \text{particle} * \text{group} \mid \text{item}) \end{aligned}$$

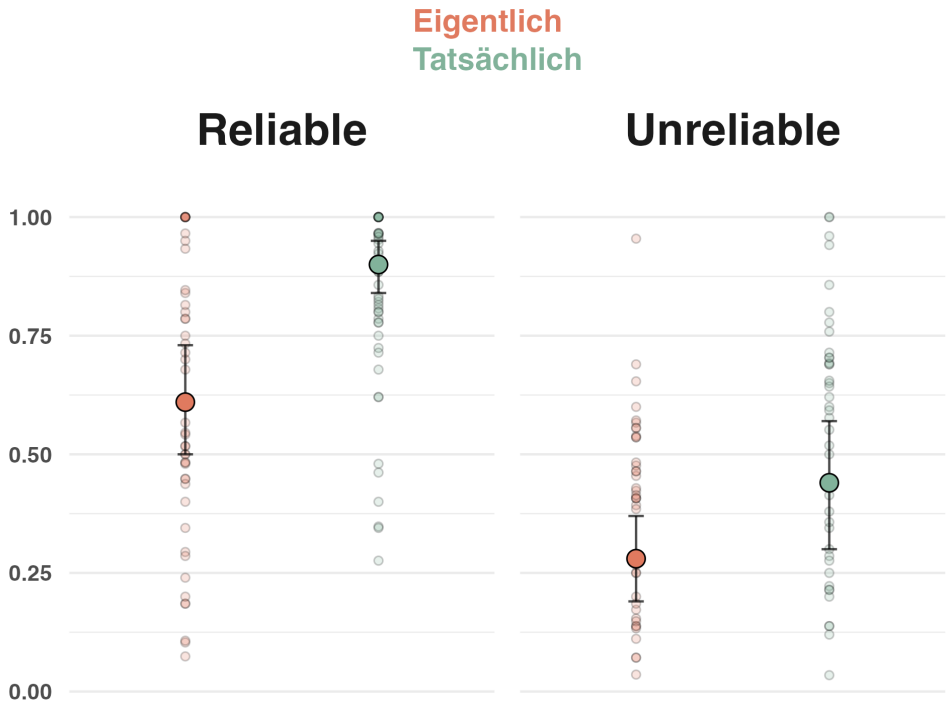

Figure 4: Mean estimated proportions of biased trials. Mean predicted proportions of biased trials (large circles) with 95% credibility intervals, overlaid to empirical participant means (small circles). Values closer to 1 on the y-axis represent a larger proportion of trials spatially biased for the target referent. The left column shows the estimates for the **reliable** group, while the right column shows the estimates for the **unreliable** group. The estimates for *tatsächlich* are shown in green while the ones for *eigentlich* are shown in orange.

In the **reliable** group, 82% of *tatsächlich* trials were found to show a predictive bias for the mentioned referent (β = 0.90, 95% CrI [0.84, 0.95]), whereas 54% of *eigentlich* trials were found to show a predictive bias for the unmentioned referent (β = 0.61, 95% CrI [0.50, 0.73]). This provides evidence for hypotheses 1a and 1b, where $\bar{M}_{rel,tat} > 0$ and $\bar{M}_{rel,eig} > 0$. The fact that the proportion of biased *eigentlich* trials is considerably smaller than that of *tatsächlich* trials ($\bar{M}_{rel,tat} > \bar{M}_{rel,eig}$), with an estimated median difference of 28 percentage points (95% CrI [0.20, 0.37]), provides evidence for hypothesis 1c.

In the **unreliable** group, 47% of *tatsächlich* trials were found to show a predictive bias for the mentioned referent (β = 0.44, 95% CrI [0.30, 0.57]), whereas 33% of *eigentlich* trials were found to show a predictive bias for the unmentioned referent (β = 0.28, 95% CrI [0.19, 0.37]). In other words, there is a considerable drop in the number of biased trials when comparing the **unreliable** group to the **reliable** group, in particular when considering *tatsächlich* trials. Notice also that there is much more uncertainty around the estimated proportion of *tatsächlich* trials in the **unreliable** group compared to the **reliable** group, as per Fig. 4. The fact that the proportions are considerably smaller in the **unreliable** group (i.e. $\bar{M}_{rel,tat} > \bar{M}_{unrel,tat}$ and $\bar{M}_{rel,eig} > \bar{M}_{unrel,eig}$), with an estimated median difference of 46 percentage points (95% CrI [0.32, 0.60]) in the case of *tatsächlich* and 34 percentage points (95% CrI [0.18, 0.48]) in the case of *eigentlich*, provides evidence for hypotheses 2a and 2b.

2.2.3 Block effects

We also explore whether the effects found in the section above vary depending on whether the trajectories come from early or late stages of the experiment, such that participants might adapt their behavior as they progress through the experimental task. As for the MDRR, in the **reliable**

group there was no considerable change in the values over time neither in the *tatsächlich* trajectories (Block 1 = 0.35; Block 10 = 0.37) nor in the *eigentlich* ones (Block 1 = 0.08; Block 10 = 0.09). In the **unreliable** group, there was a considerable decrease in the MDRR over time in the *tatsächlich* trajectories (Block 1 = 0.22; Block 10 = 0.09), whereas there was no considerable change in the *eigentlich* trajectories (Block 1 = 0.04; Block 10 = -0.01).

As for the proportion of spatially biased *tatsächlich* and *eigentlich* trials within each experimental group, as before, we fitted a Bayesian hierarchical model in order to quantify uncertainty over the reported empirical results. More specifically, we fitted an extended version of the model described in the section above, shown in `brms` syntax below. The model predicted the probability of each experimental trial being biased towards the target referent as a function of an interaction between the particles and the experimental groups, as in the previous section, but also experimental blocks (first vs. last block). We plot the proportions of biased trials as predicted by the model atop the empirical means in Figure 5.

$$\begin{aligned} \text{horiz_bias} \sim \text{particle} * \text{group} * \text{block} \\ (1 + \text{particle} * \text{block} \mid \text{participant}) + \\ (1 + \text{particle} * \text{group} * \text{block} \mid \text{item}) \end{aligned}$$

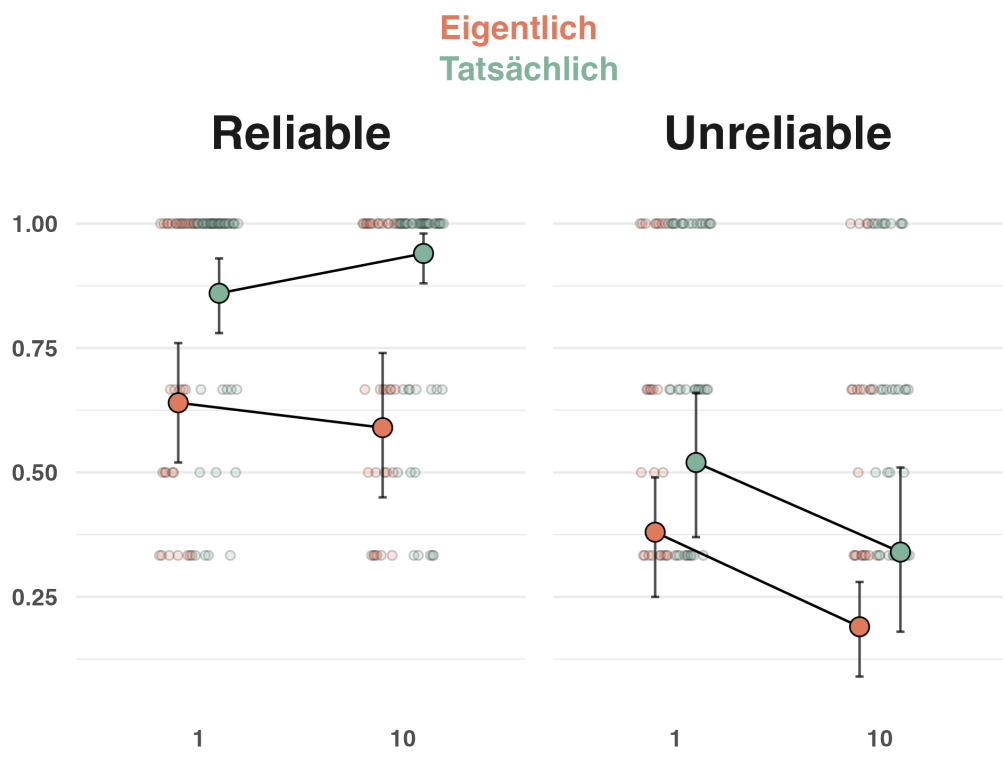

Figure 5: Mean estimated proportions of biased trials in experimental blocks 1 and 10. Comparison between mean predicted proportions of biased trials (large circles) at the beginning (experimental block 1) and end of the experiment (experimental block 10), overlaid to empirical participant means (small circles). Values closer to 1 on the y-axis represent a larger proportion of trials spatially biased for the target referent. The left column shows the estimates for the **reliable** group, while the right column shows the estimates for the **unreliable** group. The estimates for *tatsächlich* are shown in green while the ones for *eigentlich* are shown in orange.

In the **reliable** group, 78% of *tatsächlich* trials in block 1 were found to show a predictive bias for the mentioned referent (β = 0.86, 95% CrI [0.78, 0.93]), whereas 59% of *eigentlich* trials in block 1 were found to show a predictive bias for the unmentioned referent (β = 0.64, 95% CrI [0.52,

0.76]). In contrast, 80% of *tatsächlich* trials in block 10 were found to show a predictive bias for the mentioned referent (β = 0.94, 95% CrI [0.88, 0.98]), whereas 54% of *eigentlich* trials in block 10 were found to show a predictive bias for the unmentioned referent (β = 0.59, 95% CrI [0.45, 0.74]). In other words, the number of biased trials remains relatively stable when comparing the first and last experimental blocks, with an estimated median increase of 7 percentage points (95% CrI [0.01, 0.15]) in the case of *tatsächlich* trials and an estimated median decrease of 5 percentage points (95% CrI [-0.19, 0.09]) in the case of *eigentlich* trials.

In the **unreliable** group, 60% of *tatsächlich* trials were found to show a predictive bias for the mentioned referent (β = 0.52, 95% CrI [0.37, 0.66]), whereas 40% of *eigentlich* trials were found to show a predictive bias for the unmentioned referent (β = 0.38, 95% CrI [0.25, 0.49]). In contrast, 46% of *tatsächlich* trials in block 10 were found to show a predictive bias for the mentioned referent (β = 0.34, 95% CrI [0.18, 0.51]), whereas 30% of *eigentlich* trials in block 10 were found to show a predictive bias for the unmentioned referent (β = 0.19, 95% CrI [0.09, 0.28]). In other words, unlike in the **reliable** group, there is a more substantial change in the number of biased trials when comparing the first and last experimental blocks, both in the case of *tatsächlich* and *eigentlich*, with an estimated median decrease of 19 percentage points in both cases (95% $\text{CrI}_{\text{tatsächlich}}$ [-0.33, -0.03]); 95% $\text{CrI}_{\text{eigentlich}}$ [-0.31, -0.07]).

3. Discussion

In the present study, we investigated whether modal particles affect incremental interpretation and the online resolution of reference. To that end, we used a mouse-tracking task where participants were asked to listen to short dialogues which included a polar question introducing a referent in discourse followed by an aswer containing one of two German particles, either *tatsächlich* or *eigentlich*. For each dialogue, participants were presented with two images on a screen and instructed to select the one that best portrayed the topic of the exchange. Crucially, we were interested in the differential effects of processing *tatsächlich* and *eigentlich* in context, as measured in terms of a bias in the horizontal position of the mouse after hearing the particle and before hearing the critical referent. We were also interested in the effects of varying the pragmatic reliability of the particles, as manipulated between experimental groups, one including only reliable particle usages, and one including both reliable and unreliable usages.

Modal particles as predictive pragmatic cues

The results show that, in the reliable group, both *tatsächlich* and *eigentlich* were used predictively in interpretation, such that their presence in the critical sentence induced participants to move their mouse towards the relevant referential target – which was different for each particle – before hearing information from the disambiguating noun. The results also show that exposure to unreliable particle usages not only led to temporal modulations of the qualitative patterns found in the reliable group but, crucially, it reduced the predictive potential of the particles, as evidenced by an overall reduction in the proportion of trajectories biased for the relevant referent compared to the reliable group.

What the results suggest is that, as found for Dutch *inderdaad* and *eigenlijk* in the study by van Bergen and Bosker (2018), particles with a modal-marking function do affect incremental interpretation, at least in the context of reference resolution. Crucially, they affect not only the outcome of the interpretive process but also how it unfolds in real-time. While the contrast-marking particle *eigentlich* modulates interpretation insofar as it induces comprehenders to revise an assumption at issue in the discourse, e.g., that the discourse referent under consideration is actually a dog as opposed to a wolf, the agreement-marking particle *tatsächlich* modulates interpretation insofar as it induces comprehenders to confirm a prior assumption at issue in the discourse, e.g.,

that the discourse referent under consideration is indeed a dog as opposed to a relevant alternative like a wolf.

Our results also add to the body of literature showing that pragmatic prediction is prone to being modulated by the reliability of the input (e.g. Kurumada, Brown, Bibyk, et al., 2014; Roettger & Franke, 2019). In our case, it was shown that by modulating felicity by reversing the mappings between the form and meaning of *tatsächlich* and *eigentlich*, interpreters eventually adapted to unreliable input insofar as they were less likely to draw predictive inferences compared to when only being exposed to reliable input.

In terms of the interpretive processes underlying the understanding of modal particles in context, interpretation seems to depend on a pragmatic calibration between their specialized modal meaning and any other meanings they might have. Conventional implicatures, when construed as a form of pragmatic inference (Feng, 2010, 2011), can be thought to capture exactly such process, that is, the derivation of a context-sensitive interpretation by means of association between contextual constraints and the conventionalized meaning(s) of an expression. Indeed, while earlier work looking at the processing of particles suggests that the more conventionalized the modal meaning of a particle, the harder it is to process it (Dörre et al., 2018), the present findings suggest that inferential processes play a crucial role in the interpretation of modal particles in context.

Much like the findings that prosodic information can affect incremental interpretation by providing pragmatic cues in online processing (Kurumada, Brown, Bibyk, et al., 2014; Roettger & Franke, 2019), our findings illustrate that interpreters are able to pick up on the conventionalized pragmatic contribution of linguistic forms which do not necessarily affect the truth-conditional meaning of the processed input while nevertheless still impacting interpretation in real-time, in line with previous work looking at the processing of discourse markers (Asr & Demberg, 2020; Scholman et al., 2017, 2025). In particular, this reflects a similar point about the potential pragmatic contribution of particles as made by Järnefelt et al. (2025), who found that the Swedish particles *faktiskt* and *egentligen* (which have similar meaning profiles to *tatsächlich* and *eigentlich*, respectively) are processed differently, suggesting an online sensitivity to particle meaning, despite not systematically impacting offline interpretation. While Järnefelt et al. (2025) looked at the processing of particles in low-constraining contexts, unlike our setup, they too argue that interpreters draw on a combination of pragmatic expectations and linguistic knowledge to make sense of the particles in real time, construing it theoretically in terms of a reanalysis process based on Gricean quality rather than in terms of inferencing by convention.

Modal particles as specialized discourse markers

The present findings also contribute to discussions about the status of modal particles in relation to discourse markers more generally. In light of their nature as particles with the specialized function of marking epistemicity relations, the present findings illustrate that modal particles can impact the interpretation of their host utterances by (dis)confirming propositional information, underlining their modal character and potentially warranting their special status among the more general class of discourse markers. At the same time, while in the present study the materials were designed with the explicit intention of eliciting a modal interpretation of *tatsächlich* and *eigentlich*, in van Bergen and Bosker (2018), for instance, the materials allowed both a modal as well as a non-modal interpretation of *inderdaad* and *eigenlijk*. As their results regarding *eigenlijk* suggest, a particle's modal meaning might not necessarily be the most salient meaning in context, at least not when the two meanings are relatively similar to one another, as in the case of *inderdaad/ tatsächlich* and *eigenlijk/ eigentlich*, which signal context-dependent meanings which are hard to distinguish from one another. Similarly, as the results of the study by Järnefelt et al. (2025) suggest, even though interpreters seem to be generally sensitive to the meaning of particles, that may not

necessarily lead to qualitative changes in interpretation, as pragmatically-derived interpretations might be outweighed by simpler semantic ones.

However, the difference between marking a general sense of (dis)agreement between two discourse elements and (dis)agreement between two propositions might be one of degree as opposed to type. Indeed, even in the case of particles which signal arguably quite different meanings depending on their particular usage, the difference between a particle's modalized and non-modalized meaning might not be as apparent as it seems. For instance, Dörre and Trotzke (2019) argue that there is a "common semantic denominator" between the two senses of German *bloß* and *nur*, both of which index either a subjective sense of strengthening or a restriction depending on what they relate their host utterance to. In the case of, *inderdaad/ tatsächlich* and *eigenlijk/ eigentlich*, both meanings are related to the management of expectations, whether those are expectations to do with the truth of concrete propositions or expectations related to the flow of discourse and the coherence between discourse elements.

All in all, our findings provide evidence that the German particles *tatsächlich* and *eigentlich* serve as predictive cues in the incremental processing of language, allowing comprehenders to anticipate referential material in a sentence before actually processing that material. These particles function as pragmatic cues to interpretation, modulating the illocutionary force of the utterance they occur in and allowing comprehenders to hone in on intended meaning by either confirming or disconfirming assumptions about the content of that same utterance. The nature of the modulation is one which depends on the modal meaning associated with either particle, suggesting an interpretive process akin to the derivation of conventional implicatures, that is, one which builds on both conventionalized non-truth-conditional meaning as well as pragmatic expectations.

Data availability statement

The data and analysis scripts are available for inspection under https://osf.io/dt8r3/overview?view_only=55edd0e9fabf4e8d8eb3e7d375fe8437.

Ethics and consent

The study was approved by the ethics committe of University X. Participants gave their informed consent before taking part in the study. They were informed that they could leave the experiment at any point during participation, and that, upon completion, their anonymized responses would be stored and used for academic purposes.

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Competing interests

The authors declare no competing interests.

Authors' contributions

Conceptualization: A1 and A3; Data curation: A1; Formal analysis: A1, A2, and A3; Funding acquisition: A3; Investigation: A1; Methodology: A1, A2, and A3; Project administration: A1 and

A3; Resources: A1 and A3; Software: A2 and A3; Supervision: A3; Validation: A1, A2, and A3; Visualization: A1; Writing - Original draft: A1; Writing - Review & editing: A1, A2, and A3.

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